

Vision and Scope Document

for

Theia

Version 1.0 approved

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Cpts 484: Software Requirements

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1: Introduction

1.1 Purpose: The purpose of this vision document is to clearly describe the goals, scope, and expectations for Theia, a smartphone indoor navigation application designed to assist the blind and visually impaired. This document describes the vision of the project to the stakeholders involved.

1.2 Scope: Theia is a mobile application that will be functional and user-friendly. Theia must successfully guide the users to their destination. The vision document outlines the environments of different processes and other documents that are involved with Theia. Some of these include backend services, project management processes using the Agile method, user safety workflows, WRS documentation, and functional prototypes.

1.3 Definitions, acronyms, Abbreviations:

Theia: Indoor navigation application

GitHub: Version control developer platform used for tracking each team member's individual process on the project.

Visually impaired person: Any person with a visual disability, usually blind

1.4 References: WRS Document

Theia As-Is/ To-Be Presentation

GitHub Repository

Project Preliminary Plan

IBM. (n.d.). *Requirements vision document* — DOORS Next 7.0.3. IBM Engineering Lifecycle Management Suite. Retrieved November 17, 2025, from <https://www.ibm.com/docs/en/engineering-lifecycle-management-suite/doors-next/7.0.3?topic=requirements-vision-document>

1.5 Overview: The vision document allows for a higher level of understanding regarding the Theia project. Section one details the project as well as the supporting documents. Section two describes the users; section three and four describe the product overview and product features.

2: Positioning

2.1 Business Opportunity: The project addresses the need for more user-friendly apps for those with visual disabilities. There are many applications that focus mainly on outdoor navigation and lack the ability to provide reliable indoor navigation. Theia allows for an opportunity to combine the extensive technology that is present within the outdoor applications and integrate it within this application. This application is highly accessible, affordable, and user-friendly, allowing for great mobile market value.

2.2 Problem Statement: The problem is there is little to no reliable indoor navigation systems that adequately support the blind and visually impaired. The impact of the problem is ensuring safety for these individuals as there is a high-risk regarding accidents, mobility, and an increase in relying on a caretaker to safely navigate them. A successful solution is to create an accurate indoor navigation system with real time guidance and adaptability, with emergency notifications that support the user and caretaker if necessary.

2.3 Produce Position Statement: This application is for blind and visually impaired individuals that will guarantee them a way to safely navigate indoors. There will be real time guidance, adaptability, and built-in smartphone sensors to ensure quality. Our application delivers emergency support and will have a simple design allowing for easy usage.

3: Stakeholder and user descriptions

3.1 Market Demographics: The primary market is for individuals that are blind or visually impaired. Approximately 1 million people are pronounced legally blind, and over 12 million are visually impaired. Some secondary markets include caregivers, family members of those with disabilities, and other organizations that support those with visual disabilities.

3.2 Stakeholder Summary:

Name: End Users

Represents: Individuals who are blind or visually impaired will be using the Theia app directly for indoor navigation.

Role: Provide necessary feedback while using the app. Will help to shape the design, functionality and other aspects within the app.

Name: Caregivers and Family Members

Represents: Individuals who might help set up the app, or assist the visually impaired users that will be using the app.

Role: Support user safety, ensuring the emergency protocols work and testing out the app beforehand to ensure proper usability for the actual user.

Name: Development Team

Represents: Individuals who are helping to create the app, including the coding engineers and designers.

Role: Design, test, remodel and maintain a productive working system. Improve app when necessary

3.3 User Summary:

Name: Visually Impaired users

Description: Will directly interact with the Theia application for indoor navigation. They will rely on the system instructions from the sensors and real-time voice messages in the respectable language that the app provides.

Stakeholder: End users

Name: Caregiver, Support User

Description: Assists the user while setting up the app and ensuring the correct emergency information is set up, if needed by the user. Assists the user to navigate indoor spaces that they are unfamiliar with. Helps the user with the application in the beginning until they fully understand and are comfortable with using the app on their own.

Stakeholder: Caregiver and Family Members

3.4 User Environment:

People involved: There are between 1-3 people involved with the app. There is the end user, the user who is visually impaired that needs the app for assistance while walking indoors. There is the caregiver that may help assist and/or set up the app for the user. There can also be a family member to help assist the user or the caregiver to successfully set up the app.

Task Cycle Duration: A usual task cycle involves the user selecting a destination, the system accepting the destination, then performing real time guidance for the user. The main factor is how far the user is from their chosen destination, this could range from 2 minutes to 10- 15 minutes. As the user begins to become accustomed to the app and the directions this time may lessen significantly.

Environmental Constraints: Users have to rely on their cellular device being charged, ensure it has good audio, while also making sure the app features on their device.

The environment needs to support low-light and crowded, loud areas, ensuring real-time updates.

System Platforms in Use: Theia is designed for smartphone devices including IOS and Androids.

3.5 Stakeholder profiles:

Stakeholder #1: End User

Description: Visually impaired individuals who will rely on Theia for proper navigation.

Type: Casual user; non-technical

Responsibilities:

- Provide feedback, report the issues that occur when they are using the app

Success Criteria:

- The User defines success by safely reaching their destination after following the real-time navigation read to them by the application. The end user ensures that the app can be used successfully and consistently with very minimal errors.

Involvement: Helps to ensure the features work as expected in real time. The primary users for accessibility requirements.

Deliverables: Reports functional feedback for the application, ensure that the app works as intended and meets the expectations of the users.

Comments/ Issues: May face challenges getting application set up, or if they are not familiar with the technology. Application needs to be reliable with audio clarity along with real-time responses.

Stakeholder #2: Caregiver / User Support

Description: Family member, or caregiver that assists the user when setting up the first using the application

Type: Casual user, semi technical

Responsibilities: Assist the user with the initial setup, also assist during the early stages of the application to ensure the user has a firm understanding. Communicate any other issues or concerns they see with the application so that the user is aware.

Success Criteria: The user is confident using the system and can use it independently. The application remains usable for the caregiver to assist the user if needed. There should be very minimal need for the caregiver to continue assisting the user after the application is completely set up.

Deliverables: Reports any necessary feedback that they find while assisting with the application. Ensures the emergency information they enter within the app is accurate.

Comments or Issues: The caregiver might not be familiar with the cellular device used and might need to account for clear instructions on how to properly set up the application and what it needs to install for the intended user.

Stakeholder #3: Development team/ System administrators

Description: Developers responsible for maintaining the backend services and overall application reliability, manages app evolutions.

Type: Technical experts

Responsibilities: Maintain system backend services, ensure quality over all aspects within the app. Perform needed updates allowing for newly added and created features. Ensure that the navigation occurs correctly and monitors the user's feedback to allow for application enhancements. Ensure system privacy and security for user information.

Success Criteria: Application is stable and with reliable navigation performance allowing multiple users to successfully use the application. Successfully adds new features to the app without disrupting its service to the user. User's have positive things to say about the application being reliable and successfully leading them to their destination.

Deliverables: Performs system updates and documentation. Backend services such as AP. Performance reports, supports solutions for documented user feedback.

Comments or Issues: Performance optimization within the app as it is using real-time performance. Maintain system support.

Comments or Issues: App is compliant with accessibility standards, performance optimization, and allows for direct compatibility for IOS and android devices.

3.6 *User Profiles*

User Profile 1: Visually Impaired Users (End users)

Description: Individuals who are visually impaired are going to rely on Theia for indoor navigation, depending on real-time guidance and accurate instructions.

Primary Goals: Safely navigate inside buildings relying solely on the app. Receive directions that are clear and easy to follow reducing the need to rely on others around them for navigation.

Characteristics: Limited ability, or no ability to see and read visual instructions. May not be very comfortable using a cellular device.

Technical Skill Level: Low up to moderate

Key Needs: Clear and accurate guidance with consistent communication and wording techniques. Simple flow of the application allowing for easy usage. Allows user to navigate offline or with bad connection.

Limitations and Constraints: Must rely on audio elements, cannot rely on visual aspects like most apps. Unclear instructions or system delays can hinder the apps reliability.

Description: Family member, or caregiver that assists the user when setting up the first using the application

Primary Goals: Assist with application set up, ensure user information and emergency information is correct, support user with app until they are able to use it on their own.

Characteristics: May not be as technically

Technical Skill level: Moderate to high

Key Needs: Clear instructions to allow for the app to be set up properly. Test opportunity to ensure that the app works as intended before the actual user takes over.

Allow them to report issues or give feedback in a safe manner.

Limitations or Constraints: Technical ability can range between low and extremely high.

May not be present with user every time they use the app.

3.7 Key stakeholder or user needs

Need	Priority	Concerns	Curr Solution	Proposed Solution
Reliable Indoor Navigation	High	Most systems do not have indoor navigation, it also doesn't account for those that are visually impaired	Users tend to rely on others by asking them for help with directions	Theia offers real-time indoor navigation with sensors and accurate audio instructions
Clear Audio	High	Other audio instructions may be unclear and don't allow for different languages	Closed captions on different apps that require audio	Theia allows for simple and clear voice instructions in real-time for each user

Offline Navigation	High	Some buildings have little to no service and if the app relies on having service it may not work in many different situations	Users may have to ask others for help in this instance or try to navigate the building themselves and from memory	Theia allows for the same clear and accurate navigation to work on devices that are not connected to the internet
Easy Setup	Medium	Some apps have a very complicated setup for users, making it very detailed. Users who are visually impaired do not need an extremely complicated app set	Users may ask others for help such as their caregivers or family members to help them with the app	Theia's onboarding process is extremely simple and allows the caregiver, family member or visually impaired user to set up the app more easily
Emergency Options	Medium	User's are concerned with getting lost or confused during navigation	Caregivers step in if they accompany the user and help them reach their destination. Or the user can try to navigate the building from memory if they are by themselves	Theia allows for emergency contacts and alerts that allow them to safely contact others when in an emergency situation

3.8 Alternatives and competition

Google Maps

Strengths:

- Wide Range of places
- Free
- Integrates voice navigation

Weaknesses:

- No indoor navigation
- Requires a good connection
- Instructions are sometimes confusing and complex

BlindSquare

Strengths:

- Built for those that are visually impaired
- Audio cues

Weaknesses

- Requires external sources
- Cost

NavLens

Strengths:

- Provides real time updates on location and directions
- Ideal for public spaces
- Supports multiple languages

Weaknesses:

- User must hold their phone up so that the camera can capture the marker, can be challenging at times, if crowded or narrow spaces
- Continuous use of the camera drains phone battery which is needed for the user
-
- Lighting conditions can affect the apps performance when trying to capture the landscape

4: Product Overview

4.1 Product Perspective: The product is a smartphone app designed to help with real-time guidance for those who are visually impaired, used for indoor navigation. It interacts with backend services in terms of emergencies.

4.2 Summary of Capabilities

Customer Benefit	Supporting Feature
Easy Setup	Simple setup, allowing users of all technology experience the ability to set up the application
High user satisfaction	Allows users to provide feedback to ensure the app is to the highest quality for the users
Works in crowded, noisy areas	Sensor-based navigation that doesn't rely on the camera in any way
User's can navigate without service	Offline maps with directions

4.3 Assumptions and Dependencies

- User has an IOS or android cellular device to access the app on
- User's device has working audio

- Users do not need to have internet connection at all times, but having connection will help with the app process faster

4.4 Cost and Pricing

- All backend applications used are free of cost
- The labor costs include testing, and maintaining the mobile app

4.5 Licensing and Installation

- Allows for installation via cellular device
- Privacy, Security, and accessibility standards are required

5: Product Features

5.1 Feature 1. Confirmation of Destination: Users must confirm the destination before the navigation directions start

5.2 Feature 2. Real-Time Indoor Navigation: The system provides real-time instructions, telling the user to turn or notifying the user when there is an obstacles and safely guiding the user to their destination

5.3 Feature 3. Languages/Setting Customization: Allows for clear audio with multiple different languages, allows for the user to change their settings involving volume, navigation frequency and typography

5.4 Feature 4. Offline Support: The application supports offline navigation when buildings have little to no service or Wi-Fi is unavailable. The required maps are cached for offline use.

5.5 Feature 5. Simple Onboarding: The setup for the application is extremely simple allowing the care taker or user to have a smooth onboarding process, this includes emergency information and certain device permissions

5.6 Feature 6. Accessible for Users: Larger buttons, simple menus, and a less complex visual application overall, ensuring compatibility with the users

5.7 Feature 7. Emergency: Allows for emergency assistance that the user implemented during setup, allowing the user to call contacts when they are disoriented, afraid or lost while attempting to navigate indoors

5.8 Feature 8. User Profile: Allows users to change settings for the language they prefer to use, as well as audio tone, can also change emergency contacts

5.9 Feature 9. Feedback: Users can report certain issues they come across when using the application

5.10 **Feature 10.** Storing Building Maps: The application supports loading and storing maps for many different indoor environments

5.11 **Feature 11.** Multi-Route Options: The system provides more than one route option for the user, the estimated time and necessary obstacles

5.12 **Feature 12.** Secure Data: The system safely saves the user's emergency contacts and preferences while following the necessary privacy rules

5.13 **Feature 13.** Reliability: The system responds with navigation updates within a couple of seconds, with consistent guidance

5.14 **Feature 14.** Cross-Platform Support: The application works smoothly across multiple platforms

5.15 **Feature 15. Direction Notification:** Notifies the User how far they are until their next direction

5.16 **Feature 16. Analytics for Buildings:** Allows for anonymous taps that can help identify high traffic areas, common issues, or other safety hazards that the system did not initially pick up

6: Constraints

6.1 Connectivity Constraints:

- Indoor buildings/ environments may not have very robust Wi-Fi, this requires the application to have offline caching
- Updates may be delayed based on the connection quality of the user's device

6.2 Compliance Requirements

- The application must comply with the accessibility standards and privacy regulations

6.3 Safety Requirements

- Navigation directions must be accurate to ensure user safety, and not provide the user with inaccurate navigation that could result in them getting lost or hurt
- Emergency Contacts are saved accurately and can be called by the user when in trouble, also allows the user to cancel the activation if it was unintended

6.4 Environment Constraints

- Indoor environments vary largely, in terms of lighting, noise, crowds, and the amount of obstacles present, which may impact audio clarity and the detection of obstacles

6.6 Structure Dependency

- The app relies on accurate building layouts and maps that are provided to the development team

6.6 Hardware Limitations

- The application must work on mobile devices
- The application is limited by the device's current audio, sensor quality and the device's ability to connect to the network

7: Quality Ranges

7.1 Performance

- The system gives updates and can reroute the user in about 3 seconds, depending on network connection and sensor quality
- Obstacle detection must be delivered almost immediately
- Calling emergency contact must occur almost immediately as the user activates it

7.2 Robust System

- The system must still work with the basic operations even if the user's sensors fail
- The system should notify the user if it cannot create a safe route to the desired destination
- If the system crashes it should reopen in a recoverable state

7.3 Reliability

- The system should have close to immediate replies to the user about 98% of the time
- The indoor maps remain usable until the user deletes them

7.4 Security

- The user's emergency contacts must remain secure to the user only
- Any personal data the user enters must be encrypted when stored
- Only the emergency contacts that the user inputs can be notified when there is an emergency

7.5 Usability

- The application must support adjustable audio volume, navigation frequency and typography

- The onboarding process should be quick and simple for all Users and or care takers that are setting up the application

8: Precedence and priority

8.1 High Priority Features (Must Have)

- **Feature 1.** Confirmation of Destination
- **Feature 2.** Real-Time Indoor Navigation
- **Feature 4.** Offline Support
- **Feature 6.** Accessible for Users
- **Feature 7.** Emergency
- **Feature 12.** Secure Data
- **Feature 13.** Reliability
- **Feature 15.** Direction Notification

8.2 Medium Priority Features (Should have)

- **Feature 3.** Languages/Setting Customization
- **Feature 5.** Simple Onboarding
- **Feature 8.** User Profile
- **Feature 10.** Storing Building Maps
- **Feature 11.** Multi-Route Options
- **Feature 14.** Cross-Platform Support:

8.3 Low Priority Features (Would be nice to have)

- **Feature 9.** Feedback
- **Feature 16.** Analytics for Buildings:

9: Other Product Requirements

9.1 Standards

Legal and Regulatory Standards

- **EU GDPR (General Data Protection Regulation)**
 - Ensure all user data is collected and processed legally and only for specified purposes ensuring full transparency to the user
 - Obtain explicit user consent for data collection
 - Implement Privacy-By-Design: integrate data protection into the app from the start

- Encrypt sensitive data (emergency contacts, user preferences) both in transit and at rest

Communications Standards

- **WCAG 2.1 (Web Content Accessibility Guidelines)**
 - Provide transcripts for all audio content (spoken words, speaker details, and relevant sounds).
 - Maintain a simple, clear layout.
 - Ensure compatibility with mobile features like zoom, text resizing, and keyboard navigation.

Platform Compliance Standards

- Supported mobile platforms: **iOS** and **Android**.
- Maintain consistent behavior across platforms:
 - Minimum color contrast ratios (WCAG 2.1 compliance)
 - Text resizing up to 200% while allowing for proper app usage
 - Compatibility with screen readers and assistive tools

Quality and Safety Standards

- Perform accessibility and usability testing
- Ensure accurate, real-time navigation instructions to prevent accidents
- Verify emergency features allow for immediate notifications and allow the user to cancel if activated accidentally
- Offline functionality must be reliable when network is unavailable.

9.2 System Requirements

- Supported Host operating systems
- Minimum memory of about 2 GB RAM for saving certain building maps
- No companion software required

9.3 Performance Requirements

- Navigation updates: Provides real-time navigation updates with little to no delay (about 2-3 seconds)

- Obstacle Detection: Application detects obstacles and immediately notifies users when obstacles are in the path
- Emergency: Immediate emergency contact when the user activates it (with about a 1-2 second delay)
- Offline Navigation: Full working application guidance even when there is no network
- Reliability: The system works to the User's standards 98% of the time

9.4 Environmental Requirements

- User Environment
 - Works when the user has low light or is in a crowded area
 - App has to handle device limitations (battery, etc)
 - Offline maps should be accessed without connection
- Maintenance + Error handling
 - App should recover nicely if it crashes
 - Notify users if a route cannot be safely given to them
 - Emergency contact feature allows for cancellation

10: Documentation

10.1 Notes: In the updated and final application, the user is able to navigate successfully indoors. The app correctly notifies the user when there are obstacles and how to navigate around them. We ensured that the sensor could function accurately and correctly on different devices. The simple onboarding process remained the same throughout the construction of the app.

10.2 Setup: The home screen of the app will allow the user to select whether to go to navigation, settings, or emergency response. The navigation option will prompt the user for their next location and allow the user to either type the location or speak it. The app will then show the number of steps until they have to turn, as well as speak it. Once the user has made it to their destination, the app will return to the home screen. The settings option allows the user or caretaker to tailor the app towards the user. This includes increased vocal navigation for those who need it. The emergency option will ask the user if they are ok, and if there is no response within 15 seconds, the caretaker will automatically be called.

11: Appendix

11.1 Status

ID	Feature Description	Status	Benefit/Priority	Effort	Risk	Reason/Source
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F1	Confirmation of Destination: Users must confirm the destination before the navigation directions start	Incorporated	Critical	Medium	Low	User Requirement
F2	Real-Time Indoor Navigation: The system provides real-time instructions, telling the user to turn or notifying the user when there is an obstacles and safely guiding the user to their destination	Incorporated	Critical	High	High	Core Functionality for application
F3	Languages/Setting Customization: Allows for clear audio with multiple different languages, allows for the user to change their settings involving volume, navigation frequency and typography	Incorporated	Important	Medium	Low	Accessibility Requirement, User Preference
F4	Offline Support: The application supports offline navigation when buildings have little to no service or Wi-Fi is unavailable. The required maps are cached for offline use.	Approved	Important	Medium	Low	Ensures Usability in all areas
F5	Simple Onboarding: The setup for the application is extremely simple allowing the care taker or user to have a smooth onboarding process, this includes emergency information and certain device permissions	Incorporated	Critical	Low	Low	Ensure ease of setup
F6	Accessible for Users: Larger buttons, simple menus, and a less complex visual application overall, ensuring compatibility with the users	Incorporated	Critical	Medium	Low	Accessibility guidelines for visually impaired users
F7	Emergency: Allows for emergency assistance that the user implemented during setup, allowing the user to call contacts when they are disoriented, afraid or lost while attempting to navigate indoors	Incorporated	Critical	High	High	Safety/ User Requirement
F8	User Profile: Allows users to change settings for the language they prefer to use, as well as audio tone, can also change emergency contacts	Incorporated	Important	Medium	Low	User customization
F9	Feedback: Users can report certain issues they come	Proposed	Useful	Low	Low	Optional to help improve application

	across when using the application					
F10	Storing Building Maps: The application supports loading and storing maps for many different indoor environments	Approved	Important	Medium	Medium	Supports offline navigation
F11	Multi-Route Options: The system provides more than one route option for the user, the estimated time and necessary obstacles	Incorporated	Important	High	Medium	Enhances navigation flexibility
F12	Secure Data: The system safely saves the user's emergency contacts and preferences while following the necessary privacy rules	Incorporated	Critical	Medium	Low	Privacy Standards / Compliance requirement
F13	Reliability: The system responds with navigation updates within a couple of seconds, with consistent guidance	Incorporated	Critical	High	Low	Performance Requirement, helps ensure maximum usability
F14	Cross-Platform Support: The application works smoothly across multiple platforms	Incorporated	Important	Medium	Medium	Ensures user accessibility
F15	Direction Notification: Notifies the User how far they are until their next direction	Incorporated	Critical	Medium	Medium	Core navigation requirement, improves user awareness and reduces errors
F16	Analytics for Buildings: Allows for anonymous tips that can help identify high traffic areas, common issues, or other safety hazards that the system did not initially pick up	Proposed	Useful	Low	Low	Optional Feature to improve the system performance and user safety

11.3 Effort:

This is an estimation of the efforts required to implement each feature to help with the scope of the project while also setting realistic expectations. The features with high effort will serve as the main starting point for the app because they are believed to take the most amount of time to complete. High effort features, such as sensor integration, real-time processing and audio updates, are critical to implement to ensure the application functions are at the highest standard. Features with low effort, like simple UI changes, will be focused later on in the development process.

11.4 Risk: The risk level indicates the likelihood of delays, or failure to deliver a feature within the application. Features that depend on external sources, such as backend

applications or newer functions, have a higher risk. In contrast, most standard UI operation features are typically low risk.

11.5 Stability: Measures how likely a feature's requirements or understanding may change during development. Features such as destination confirmation are well defined and are not likely to change; these features are highly stable. Whereas features involving environmental feedback, new data, or analytics have a lower stability and are more likely to change because requirements involve safety issues.